

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (Credit System) Degree Examinations
January 2023

19BT8X42 – SOLID WASTE MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain Volume reduction and Recycling opportunities in handling of Solid Waste.	10	L*5	1	1
b)	Write a note on Hauled Container System (HCS) and Stationary Container System (SCS) of collection of solid waste.	10	L2	2	1,2
2. a)	What are the requirements of vehicles used for hauling on highways?	10	L1	2	1,6
b)	Explain (i) Shredders (ii) Density Separation (iii) Conveyors.	10	L2	2	1,9
3. a)	Define Integrated Solid Waste Management and explain the activities associated with it.	10	L1	1	1,9
b)	Write a note on Transfer stations and its classification.	10	L3	2	6,7
Unit – II					
4. a)	Explain with respect to sanitary landfill (i) Cell (ii) Secure landfill (iii) Bench (iv) Landfill liner.	10	L2	4	2,7
b)	Compare Aerobic composting and Anaerobic digestion process of MSW.	10	L4	3	2
5. a)	Write a note on combustion systems used for processing of solid wastes.	10	L3	4	7
b)	Explain Anaerobic biological transformation of solid waste.	10	L3	3	2,9
6. a)	Explain any five landfill siting considerations.	10	L2	4	9
b)	Write on any five important design consideration for aerobic composting process.	10	L2	3	2,9
Unit – III					
7. a)	Explain the framework for planning activities of solid waste management.	10	L2	5	1,12
b)	Give any five duties of solid waste generators and CPCB as per SWM Rules-2016.	10	L2	5	1,9
8. a)	Give any five options / requirements for setting up of E-waste collection centres as per E-waste rule 2011.	10	L2	5	12
b)	What is Hazardous waste? Give its classification as per Hazardous waste management Rules – 2016.	10	L2	5	1,9

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
